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RESEARCH ARTICLE

Discovery of a Thrombin-Targeting Anticoagulant Peptide From *Whitmania pigra* via a "Computation-Guided Experimentation" Strategy

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Keywords: anticoagulation | peptide screening | thrombin | *Whitmania pigra*

ABSTRACT

Targeting thrombin to screen safe thrombin inhibitors from natural plants and animals is a critical direction in anticoagulant drug development. This study aimed to screen thrombin inhibitors from the nonbloodsucking leech *Whitmania pigra* (WP) and elucidate the mechanism of anticoagulation through a "computation-guided experimentation" strategy. A peptide library was constructed from WP hydrolysates, and virtual screening was performed using molecular docking and dynamics simulations. A novel thrombin-targeting anticoagulant peptide PEP^{WP} (LRELEDALEQER) was screened out from the peptide library and validated through in vitro/in vivo experiments. PEP^{WP} significantly prolonged thrombin time (TT) and prothrombin time (PT) in a dose-dependent manner in vitro, indicating its role in the common and extrinsic coagulation pathways. Surface plasmon resonance (SPR) analysis then confirmed strong thrombin binding ($K_d = 7.242 \times 10^{-6}$ mol/L). Furthermore, PEP^{WP} prolonged TT while reducing blood viscosity in acute blood stasis rats. Finally, structural analysis revealed that PEP^{WP} bound to Exosite II of thrombin. Arg233 and Arg101 were the key residues for the binding. In conclusion, PEP^{WP} exhibited good anticoagulant activity and significant application potential.

1 | Introduction

Cardiovascular diseases (CVDs) are the main cause of mortality and morbidity worldwide, leading to nearly 420 million diagnoses and 17.9 million deaths annually, and the death number is expected to exceed 23.6 million by 2030 [1, 2]. As a core strategy for thrombosis prevention, anticoagulation plays a key role in the prevention of CVDs by targeting and inhibiting the coagulation cascade.

Thrombin, as the terminal effector serine protease in the coagulation cascade, plays a central and dual role in cardiovascular pathophysiology. Upon endothelial injury, thrombin initiates thrombus formation through proteolytic cleavage of

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fibrinogen. As the primary driver of thrombosis, it is also a key target of antithrombotic therapy. Structurally, thrombin's catalytic machinery comprises two exosites (Exosites I and II) governing substrate recognition and a catalytic triad (Ser195, His57, and Asp102) that mediates peptide bond hydrolysis [3]. These three distinct functional domains represent primary pharmacological targets for anticoagulant drug design [4, 5]. The therapeutic evolution of direct thrombin inhibitors (DTIs) reveals critical pharmacodynamic–pharmacokinetic trade-offs across generations. First-generation DTIs, exemplified by hirudin, achieve potent anticoagulation through bivalent binding to the catalytic Ser195 residue and Exosite I, inducing irreversible thrombin inactivation. This noncompetitive inhibition mechanism results in significant bleeding liabilities without specific antidotes, necessitating strict hemostasis monitoring, which limits the clinical utility of hirudin [6, 7]. Second-generation synthetic analogs like bivalirudin demonstrate improved safety profiles through rationally designed peptide bonds that enable rapid renal clearance ($t_{1/2} = 25$ min). However, their reversible exosite-targeted binding still carries a 3% incidence of major bleeding [8]. The third-generation small-molecule DTIs, represented by Argatroban and Dabigatran, which can selectively bind to the catalytic triad, show a paradigm shift in target specificity. They demonstrate a better safety profile compared to first- and second-generation DTIs. Whereas the former requires continuous intravenous infusion due to its short half-life, the latter, as an oral direct thrombin inhibitor, does not require routine INR monitoring but exhibits a > 50% reduction in clearance in patients with renal impairment [9].

Accumulating evidence demonstrates that allosteric regulation serves as a fundamental mechanism for thrombin functional modulation, exerting its effects through allosteric control of the cascading amplification of thrombin inhibition [10]. Studies have revealed that the α-helix region within Exosite II represents a critical allosteric regulatory site. This region induces spatial misalignment of the catalytic triad, leading to contraction of the substrate-binding pocket [11]. Concurrently, it destabilizes the conformational stability of thrombin's fibrinogen-binding domain (Exosite I), significantly reducing fibrinogen affinity. Furthermore, by interfering with thrombin's interaction with the PAR-1 receptor, this mechanism suppresses platelet activation signaling pathways [12]. For example, a synthetic sulfated lignin that binds Exosite II not only modulated thrombin activity but also attenuated thrombosis without increasing bleeding in mouse thrombosis and bleeding models [13]. These findings make Exosite II a promising new target for the development of thrombin inhibitors.

Whitmania pigra (WP), a representative species of the Hirudinidae family, has been used in Traditional Chinese Medicine (TCM) pharmacopeias for over two millennia. Research revealed that its raw or processed forms exhibited significant effects in treating CVDs such as coronary heart disease and cerebral infarction in clinical practice [14]. WP demonstrated dual advantages in clinical applications, exhibiting robust therapeutic efficacy coupled with an exceptional safety profile. No published literature reported the adverse effects, including hemorrhagic complications or hepatotoxicity/nephrotoxicity. This remarkable safety margin compared with conventional antithrombotic agents makes WP a promising candidate for long-term therapeutic applications. Some anticoagulant peptides have been isolated from WP, but their targets and anticoagulant mechanism were still unclear [15–17]. The acquisition of active peptides by traditional separation methods, such as ion-exchange chromatography and gel chromatography, is often accompanied by a long separation term and high economic costs, which have limited the discovery of thrombin inhibitors from TCM.

Given the limitations of traditional drug screening methods, computer-guided drug screening has emerged as an indispensable tool in modern drug discovery due to its high efficiency and high-throughput capabilities. Among these technologies, virtual screening stands out as a particularly prominent approach, primarily encompassing two core methodologies: molecular docking and molecular dynamics (MD) simulations. These techniques can be employed either independently or synergistically: Molecular docking rapidly predicts the static binding conformations of ligand-receptor complexes, whereas MD simulations track atomic trajectories to elucidate the dynamic evolution of molecular interactions. This static-dynamic integration strategy provides a more comprehensive and reliable theoretical foundation for drug discovery [18–22].

In this study, we applied "computation-guided experimentation" approaches to screen thrombin inhibitors from WP. Given that WP is administered orally and its bioactive proteins/peptides are susceptible to degradation by gastrointestinal proteases, and our earlier work demonstrated that trypsin hydrolysis enhances bioactive peptide release—both in raw and processed WP [23]. We first performed trypsin digestion on both raw and processed WP. Subsequent mass spectrometry–based identification of tryptic digest products enabled the establishment of a comprehensive WP peptide library. Through molecular docking and MD simulations, a novel unreported peptide (PEP^{WP}: LRELEDALEQER) was screened out from this library. PEP^{WP} was verified to have anticoagulant activity by in vitro and in vivo experiments. Furthermore, our findings revealed that the binding site of PEP^{WP} to thrombin was different from that of hirudin. This discovery may offer new therapeutic potential for CVDs and establish an efficient screening strategy for thrombin inhibitors.

2 | Materials and Methods

2.1 | Enzymatic Hydrolysis of Raw and Processed WP and UPLC-Q-Exactive-MS Analysis for Enzymatic Hydrolysate of WP

To obtain more kinds of peptides, the enzymatic hydrolysis of raw and processed WP was carried out under two conditions: Condition 1, 3% enzyme to substrate ratio at 37°C for 6 h, and Condition 2, 10% enzyme to substrate ratio at 37°C for 10 h.

The MS/MS experiments were executed on a Thermo Q Exactive Focus mass spectrometer (Thermo Fisher Scientific) with an electrospray ionization (ESI) interface. Chromatographic separation was performed on a UPLC Ultimate 3000 system (Thermo

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Fisher Scientific) equipped with an ACQUITY UPLC BEH C18 column (2.1 mm × 100 mm, 1.8 μm, Waters).

UPLC chromatographic condition: The mobile phase was 0.1% formic acid-water (Liquid A) and acetonitrile (Liquid B, Thermo Fisher). An increasing linear gradient was used as follows: 0–5 min, 5%–20% B; 5–8 min, 20%–40% B; 8–20 min, 40% B; 20–22 min, 40%–90% B; 22–23 min, 90%–99% B; 23–25 min, 99%–5% B; and 25–30 min, 5% B. The flow rate was 0.3 mL min^{-1}, the column temperature was 40°C, and the injection volume was 2 μL.

2.2 | Peptide Identification

The mass spectrometry data from UPLC-Q-Exactive-MS were spectra analyzed using Qual Browser of Xcalibur 3.0.63 (Thermo Fisher Scientific), and then, the raw data were searched using Maxquant (Version 1.6.17.0). The hirudinea Uniprot database (26,495 protein sequences, downloaded on July 25, 2021) was used as the search database. The parameters were set as follows: The minimum peptide length was set to 3; the digestion method was set to Trpsin/p; the maximum missed cutting site was 2; the fixed modification was set to Carbamidomethyl (C); the variable modification settings were Acetyl (protein N-term), oxidation (M); precursor tolerance was 10 ppm; fragment tolerance was 20 mmu; the false-positive discovery rate (FDR) was set to 1%; the quantitative method was LFQ (label-free quantification), Match Between Runs; and other parameters are default [24].

2.3 | Thrombin-Peptide Docking

The docking study was based on the structure of thrombin extracted from RCSB-PDB (https://www.rcsb.org/structure/2BVR). Thrombin-peptide docking was performed with both HPEPDOCK Server (http://huanglab.phys.hust.edu.cn) and Cdocker in Discovery Studio 2016 [25]. HPEPDOCK Server was used for global flexible molecular docking, and Cdocker was used to perform semiflexible docking using hirudin TYS18 residue as the active site. Then, peptides were screened according to the docking scores for MD simulation.

2.4 | MD Simulation

The protein structure of thrombin was obtained from the Protein Data Bank database (https://www.rcsb.org/structure/2BVR). The structure of the WP-derived peptides was constructed by CHARMM-GUI (https://charmm-gui.org/) [26, 27]. MD simulations were all carried out using software GROMACS 2018.33 [28, 29], and the TIP3P water model was applied for solvent representation [30, 31]. The steepest descent method was used to simulate and optimize the system to reduce unreasonable contact or atomic overlap in the entire system; then, the restricted MD simulation was performed to balance the system under the canonical and isothermal-isobaric ensemble, making the simulation system fully prebalanced. The Verlet leapfrog algorithm was used to solve Newton's equation of motion, and the integration step was set to 2 fs. In the calculation process, the Lennard–Jones function was plotted to calculate the van der Waals (VDW) force, and the nonbond cutoff distance was set to 1.2 nm; the LINCS algorithm was employed to constrain bonds involving hydrogen atoms, and the particle mesh Ewald (PME) method was used to calculate the short-range electrostatic interaction. Periodic boundary conditions were used in the simulation. All MD simulations were performed under the isothermal and isostatic ensemble, with a temperature of 310.15 K and a pressure of 1 atm. The temperature and pressure were controlled by the V-rescale and Parrinello–Rahman methods. The temperature and pressure coupling constants were 0.1 and 0.5 ps, respectively. The MD simulation duration was set to 50 ns, and three independent production samples were performed. Visualization and graphics were generated with PyMOL software (https://pymol.org/2/support.html).

2.5 | MM/PBSA

The binding free energy of the thrombin-peptide complex was computed using the molecular mechanics/Poisson Boltzmann surface area (MM/PBSA) approach. The calculation of MM/PBSA was based on the results of the MD simulation. The binding energy is calculated at 100-ps-long intervals sampled over a 50-ns time period, generating a total of 500 frames from the trace.

2.6 | Peptide Solid-Phase Synthesis and Purity Verification

The peptides screened by molecular simulation were solid-phase synthesized by GL Biochem (Shanghai) Ltd. High performance liquid chromatography (HPLC) was used to verify the purity of the peptides. The mobile phase was 0.1% trifluoroacetic acid acetonitrile solution (A)–0.1% trifluoroacetic acid aqueous solution (B) with gradient elution (0–25 min, 15%–43%A) at a flow rate of 1.0 mL/min, the detection wavelength was 220 nm, the injection volume was 5 μL, and the chromatographic column was ChromCore 120 C18 reversed-phase column (5 μm, 4.6 × 250 mm). Mass spectrometry was utilized to determine the relative molecular mass of the peptide. The mass spectrometry detection conditions were ESI ionization source, ion source voltage +4.5 kV, and mobile phase 50% water/50% acetonitrile at a flow rate of 0.2 mL/min. The detector voltage was 1.2 kV.

2.7 | Anticoagulant Activity Assay In Vitro

The screened WP antithrombin peptide was synthesized and identified for purity. Coagulation times: activated partial thromboplastin time (APTT), prothrombin time (PT), and thrombin time (TT) were determined in human plasma (Hongquan, HQ70087) using a semiautomatic clotting analyzer (steellex) according to the manufacturer's instructions (steellex), and deionized water was used as a blank control [32]. In order to measure the coagulation time at different concentrations of the screened peptide and ensure the same volume of the coagulation system, a series of concentrations of

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the screened peptide (1, 5, 10, 15, and 20 mg mL^{-1}) were prepared using water.

2.8 | Thrombin-Catalyzed Coagulation of Fibrinogen

A 0.4% fibrinogen solution (70 μL) and different concentrations of the peptide (8, 10, and 15 mg mL^{-1}; 20 μL) were added into each well. After 10 min incubation at 37°C, the absorbance was taken at 405 nm as the "sample blank." Then, 10 μL thrombin solution (10 U mL^{-1}) was added to the wells to start the reaction of thrombin-catalyzed coagulation of fibrinogen, and absorbance was monitored at 405 nm every 20 s for 10 min at 37°C [33]. The fibrinogen was dissolved by Tris–HCl buffer (pH 7.2, 0.05 M) containing 0.12 mM NaCl; the peptide and thrombin were dissolved by water.

2.9 | Chromogenic Substrate Method

One-hundred-microliter buffer solution (0.05 mol L^{-1} Tris–HCl, pH = 8.3), 10 μL with an activity of 5 bp thrombin (dissolved in normal saline), and 20 μL anticoagulant polypeptide with different concentrations (normal saline in the blank group) were added to each well. After 3 min of incubation at 37°C, 50 μL s-2238 (1.25 mg·mL^{-1}) was immediately added to each well, and after the plate was vibrated for 10 s, the absorbance was measured every 20 s and monitored for 10 min to investigate the effects of different concentrations of peptide on thrombin.

2.10 | Surface Plasmon Resonance (SPR)

The 10 × PBS-P + buffer was diluted 10 times to prepare 1 × PBS-P + buffer for use. The thrombin was dissolved in 1 × PBS-P + buffer to achieve a concentration of 200 μg mL^{-1}. The thrombin solution was prepared into 40 μg mL^{-1} with sodium acetate at pH 4.0. Thrombin was fixed to the surface of the CM5 chip by running an amino coupling reagent (EDC/NHS = 1:1) and 1 × PBS-P + buffer at 25°C. The peptide solution was diluted to a maximum injection concentration of 100 μmol L^{-1}, and samples for gradient concentration analysis of the peptide were prepared. The injection analysis sample was set to bind to the chip for 60 s, dissociate for 60 s, and run at a flow rate of 30 μL min^{-1}. After operation, peptide-thrombin affinity was determined using the Biacore T200's built-in kinetics/affinity program.

2.11 | Animals

Twenty-four SD rats, weighing 230 g ± 20 g, male, and 30 SD rats, weighing 320 g ± 20 g, male, both purchased from Spafo (Beijing) Biotechnology Co. Ltd., Animal License No. SCXK (Beijing) 2019-0010, meeting the national standard for first-class healthy animals. The animal-related experiments involved in this study met the ethical requirements of experimental animals and were approved by the Animal Ethics Committee of Beijing University of Chinese Medicine (Approval Number: BUCM-2023101002-4019).

2.12 | Effect of PEP^{WP} on Blood Coagulation in Rats

Twenty-four SD rats were fed for 7 days and randomly divided into a blank control group, a positive control group, a high-dose peptide group, and a low-dose peptide group, with six rats in each group. The dose of peptide was 40 mg kg^{-1} in the high-dose group, 20 mg kg^{-1} in the low-dose group, and 1 mg kg^{-1} in the positive control group, all of which were administered by one-time tail vein injection, and the same amount of normal saline was injected into the blank control group. After injection, the rats were briefly anesthetized with isoflurane at 5, 15, 30, 60, and 90 min. The whole blood was anticoagulated with 1:9 sodium citrate and then centrifuged at 3000 rpm for 10 min, and plasma was obtained for use. The TT, PT, APTT, and fibrinogen time of each group of rats at different action times after administration were measured by an automatic coagulation instrument, and each sample was measured once.

2.13 | Effect of PEP^{WP} on Hemorheology of Acute Blood Stasis Rats

Thirty SD rats were fed for 7 days and randomly divided into a blank control group, a positive control group, a high-dose peptide group, and a low-dose peptide group, with six rats in each group. The dose of peptide was 40 mg kg^{-1} in the high-dose group, 20 mg kg^{-1} in the low-dose group, and 1 mg kg^{-1} in the positive control group, all of which were administered by one-time tail vein injection, and the same amount of normal saline was injected into the blank control group. Fifteen minutes after injection, except for the blank control group, the other groups were injected with 300 mg kg^{-1} high-macromolecule dextran solution through the tail vein to establish the acute blood stasis model of rats, and the blank control group was injected with the same volume of normal saline. After 60 min of modeling, blood was taken from the abdominal aorta, and the whole blood was removed and stored in a heparin sodium anticoagulant tube. The whole blood was tested for hemorheological indexes.

2.14 | Statistical Analysis and Visualization

Statistical analysis and graphical representations were performed using Graphpad Prism 6.02 software. One-way analysis of variance (one-way ANOVA) was used, and $p < 0.05$ was considered statistically significant.

3 | Results

3.1 | One-Hundred-Sixty Peptides of WP Were Identified

The enzymatic hydrolysate of WP was analyzed by UPLC-Q-Exactive-MS mass spectrometry (Figures S1–S4). The enzymatic hydrolysates were identified by Maxquant in positive ion mode, and then, potential contaminants and peptides with an identification score of less than 50 were removed. The species analysis of peptides in the two enzymatic hydrolysates was performed; a

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total of 160 distinct peptides were identified from the two enzymatic hydrolysates.

3.2 | Molecular Docking Screened out Nine Peptides With High Thrombin-Peptide Docking Scores

Molecular docking can explore the conformational space, show how proteins interact with ligands, and predict the binding affinity of ligand-protein. The performance of each molecular docking software is different, and the docking results will have a large deviation. In order to increase the probability of screening out anticoagulant peptides, we chose two molecular docking methods. HPEPDOCK Server and Discovery Studio Client 2016 were used for docking. Both raw and processed WP were used in clinical practice; in order to obtain more candidate peptides, all the identified peptides in four enzymatic hydrolysates were used to perform thrombin-peptide docking. Hirudin polypeptide fragment (GFFEEIPEEYLQ) was selected as a positive control. Using HPEPDOCK Server for global docking, it was found that most docking scores were less than −160, which suggested that the global flexible docking site is more selective and the possibility of successful docking is greater.

According to the results of HPEPDOCK and literature reports, TYS18 in hirudin fragments was defined as the active site, and semiflexible docking (CDocker) was performed by Discovery Studio for peptides with high global docking scores. As a result, nine peptides whose docking scores were greater than the control peptide were screened out (shown in Table 1) for the next MD study to investigate their binding stability, and the total result of docking was shown in Table S1.

TABLE 1 | Results of thrombin-peptide docking.

No.	Sequence	Length	HPEPDOCK score	CDocker_Energy score	CDocker_interaction_Energy score
Control	GDFEEIPEEYLQ	12	−206.381	181.713	127.887
1	DSYVGDEAQSK	11	−164.064	196.941	93.853
2	DSYVGDEAQSKR	12	−185.905	181.738	84.530
3	GILAADESTATIGK	14	−168.670	185.531	100.144
4	LRELEDALEQER	12	−168.419	194.252	112.741
5	AVFVDLEPTVVDEVR	15	−188.442	184.941	115.911
6	GEEVTVTTESMEEK	14	−170.361	219.695	133.149
7	QLGNAESDVGALTR	14	−211.557	200.160	103.412
8	QLLEADEVINITLSK	15	−170.458	216.714	124.825
9	TLEQSEQEALQR	12	−170.805	206.625	124.845

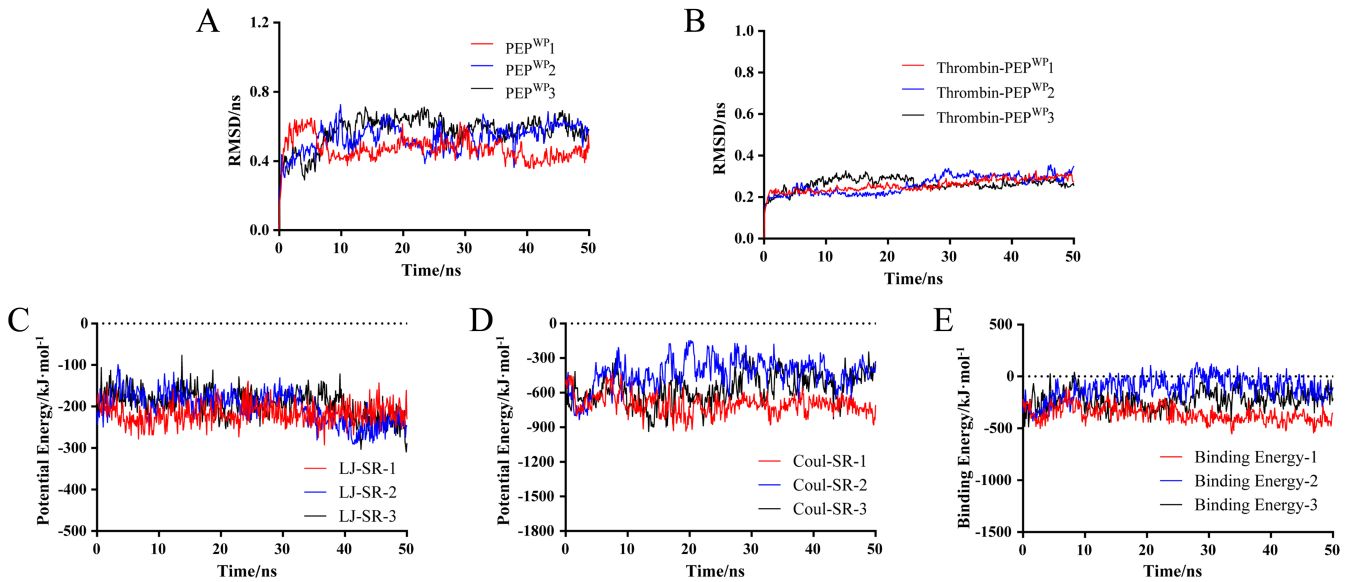


FIGURE 1 | (A) RMSD profiles of the PEP^{WP} during 50-ns MD simulation. (B) RMSD profiles of the thrombin-PEP^{WP} complex during 50-ns MD simulation. (C) Lennard–Jones potential energy analysis of the thrombin-PEP^{WP} complex. (D) Coulomb potential energy analysis of the thrombin-PEP^{WP} complex. (E) The total binding free energy of thrombin and PEP^{WP} in 500 frames of samples in 50-ns simulated trajectories. Three independent samples were performed.

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3.3 | MD Simulation and MM/PBSA Analysis of Thrombin-Peptide Complex

MD simulation was a useful tool to analyze the interaction and binding stability of the protein-peptide complex, and MD simulations provide valuable structural and energetic information of our docking models of the thrombin-peptide complex. To investigate the effect of the screened peptide on thrombin, three independent simulations (50 ns) of the thrombin-peptide complex were carried out for each peptide screened out of the previous molecular docking results (Table S2).

The dynamic stability of the thrombin-peptide complex was analyzed by studying the backbone RMSD (root mean square deviations) from the initial docked structures throughout the entire trajectory of 50 ns. Among all the peptides, the sequence of LRELEDALEQER achieved the most stable RMSD (Figure 1A,B).

The MM/PBSA was used to perform binding free energy calculations and energy decomposition for conventional MD trajectories. The binding energy was composed of van der Waals energy, electrostatic energy, polar solvation energy, Solvent-Accessible Surface Area energy (SASA), and polar solvation energy. In order to calculate the binding energy of the thrombin-peptide complex, MM/PBSA calculations were performed on all nine complexes screened by molecular docking for MD simulations; the total result was shown in Table 2. All nine complexes possessed a negative free binding energy; among them, the peptide (LRELEDALEQER) had the lowest binding energy of $-318 \pm 120.914\,kJ\,mol^{-1}$, and it was named PEP^{WP} peptide. The total binding energy of thrombin-PEP^{WP} in 500 frames was plotted, and the result was shown in Figure 1E. It was found that the total binding energy of the complex fluctuated slightly within 50 ns, indicating that PEP^{WP} and thrombin bind relatively stable within the 50-ns simulation. Coulomb potential of the thrombin-PEP^{WP} complex was around $-584 \pm 155.393\,kJ\,mol^{-1}$, much lower than the Lennard-Jones potential, which was stable at $-163 \pm 49.836\,kJ\,mol^{-1}$ throughout the simulation (Figure 1C,D), suggesting that electrostatic force was the driving force for the binding of thrombin-PEP^{WP} complex.

The thrombin-PEP^{WP} interaction was visualized using LigPlot (Figure 2). As shown in Figure 2A,B, hydrogen bonds and hydrophobic contacts between nonpolar residues were observed in the thrombin-PEP^{WP} complex both before and after molecular dynamics (MD) simulation. Notably, the hydrogen bonds between residues Arg233, Arg101, and PEP^{WP} remained stable after MD simulation, suggesting that Arg233 and Arg101 play a more critical role in the interaction between thrombin and PEP^{WP}. From Figure 2C, PEP^{WP} was found to bind to the active groove of thrombin, adopting a partial α-helical conformation. Based on the distances between Arg233-Asp6, Arg101-Glu11, and Glu127-Arg2, salt bridges likely exist between these three pairs of amino acid residues.

The binding energy and interaction analysis revealed that salt bridges and hydrogen bonds contribute significantly to the Coulomb potential energy, serving as the primary driving force for thrombin-PEP^{WP} complex formation. In addition, hydrophobic contacts between nonpolar residues also contribute to stabilizing its structure.

TABLE 2 | Results of MM/PBSA of nine screened peptides.

No.	Sequence	van der Waals energy ($kJ\,mol^{-1}$)	Electrostatic energy ($kJ\,mol^{-1}$)	Polar solvation energy ($kJ\,mol^{-1}$)	SASA energy ($kJ\,mol^{-1}$)	Binding energy ($kJ\,mol^{-1}$)
1	DSYVGDEAQSK	−232.428 ± 22.775	−1531.481 ± 212.602	1641.778 ± 169.651	−28.906 ± 2.303	−151.037 ± 82.410
2	DSYVGDEAQSKR	−68.652 ± 37.670	−1470.459 ± 450.169	1370.834 ± 393.903	−14.335 ± 5.794	−182.612 ± 114.679
3	GILAADESTATIGK	−210.745 ± 43.595	−1235.573 ± 234.008	1321.524 ± 218.666	−29.432 ± 4.901	−154.226 ± 80.896
4	LRELEDALEQER	−115.103 ± 35.317	−3114.720 ± 494.470	2933.266 ± 441.736	−22.251 ± 4.863	−318.809 ± 120.914
5	AVFVDLEPTVVDEVR	−293.924 ± 21.934	−914.869 ± 182.591	1219.490 ± 175.123	−31.622 ± 2.808	−20.925 ± 81.381
6	GEEVTVTTESMEEK	−197.335 ± 35.762	−2957.441 ± 334.927	2901.965 ± 294.934	−31.675 ± 4.396	−284.486 ± 115.430
7	QLGNAESDVGALTR	−250.921 ± 36.464	−1700.010 ± 232.301	1807.586 ± 197.073	−36.920 ± 4.222	−180.265 ± 79.247
8	QLLEADEVINITLSK	−186.860 ± 39.206	−1570.484 ± 316.032	1571.105 ± 266.748	−28.441 ± 5.387	−214.680 ± 125.777
9	TLEQSEQEALQR	−190.986 ± 37.860	−2583.162 ± 291.562	2502.047 ± 252.341	−30.372 ± 4.366	−302.473 ± 103.083

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3.4 | Solid-Phase Synthesis and Purity Verification of PEP^{WP}

PEP^{WP} was synthesized using the Fmoc solid-phase peptide synthesis method. Its purity and molecular weight of the anticoagulant peptide were determined by high-performance liquid chromatography (HPLC) and mass spectrometry (Figure S5). Mass spectrometry analysis confirmed that the molecular weight of the synthesized peptide was 1500.61, which matches the molecular weight of the anticoagulant peptide LR. The content of the anticoagulant peptide was calculated based on the peak area in the HPLC chromatogram, yielding a purity of 98.36%, indicating a high-purity peptide.

3.5 | PEP^{WP} Had Anticoagulant Activity In Vitro

The blood coagulation process is composed of the endogenous pathway, exogenous pathway, and the common pathway. When the coagulation process develops to the point

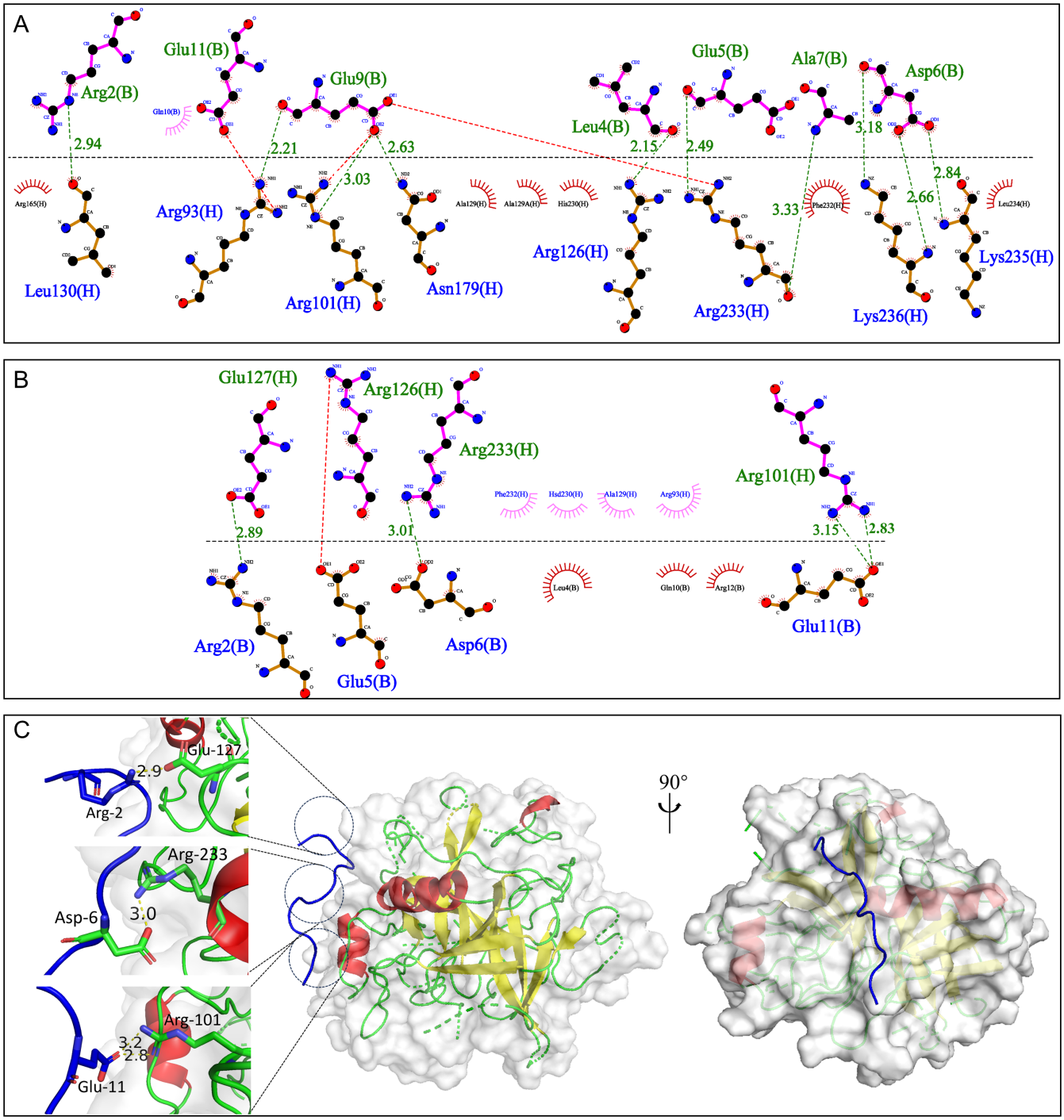


FIGURE 2 | (A) 2D mapping of the interaction between PEP^{WP} and thrombin of the docked complex by using Ligplot software. (B) 2D mapping of the interaction between PEP^{WP} and thrombin after MD simulation by using Ligplot software. (C) 3D mapping of thrombin-PEP^{WP} complex after MD simulation. Green dashed lines represent hydrogen bonds.

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where factor X is activated through exogenous and endogenous pathways, the two pathways are combined into a common pathway. In the clinical area, APTT, PT, and TT are used to evaluate the function of the above-mentioned three pathways. The anticoagulant activity of PEP^{WP} was evaluated by APTT, PT, and TT. Compared with the control buffer, PEP^{WP} could prolong TT significantly. The prolonged effect on TT was dose dependent. An obvious difference had been observed at 0.5 mg mL^{-1} in the system (Figure 3A), which indicated that PEP^{WP} could act on the common coagulation pathway. Also, PEP^{WP} prolonged the PT in plasma in a dose-dependent manner, and a significant difference was observed at 0.67 mg mL^{-1} in the system (Figure 3B), which indicated that PEP^{WP} could also play an inhibitory role in the exogenous coagulation pathway. Furthermore, PEP^{WP} did not affect the APTT (Figure 3C), showing that the endogenous pathway was not inhibited.

3.6 | PEP^{WP} Inhibited Thrombin-Catalyzed Coagulation of Fibrinogen

Thrombin catalyzes the conversion of fibrinogen to fibrin, which in turn causes thrombosis, similar to TT. Therefore, the effect of PEP^{WP} on thrombin-catalyzed coagulation of fibrinogen was studied. Under normal circumstances, fibrinogen was catalyzed by thrombin and gradually solidified; the light transmittance of the system decreased; and the absorbance increased and finally remained at a stable value. If fibrinogen was inhibited, the light transmittance of the system increased and the absorbance decreased. As shown in Figure 3D, fibrinogen coagulation can be almost completely inhibited at 3 mg mL^{-1} of PEP^{WP} in the system, and the rate of inhibition of fibrinogen coagulation at 2 mg mL^{-1} in PEP^{WP} in the system was significantly lower than that in the blank group.

3.7 | PEP^{WP} Inhibited Thrombin Activity

There are many methods to determine the binding of protein and peptide. For coagulation factors, the chromogenic substrate method is a faster and more accurate chemical detection method. As Figure 4 showed, the activity of thrombin increased continuously within 3 min after the addition of s-2238 and remained unchanged 3 min later. Compared with the blank group, 20 mg mL^{-1} of PEP^{WP} could slow down the increase rate of thrombin activity and finally downregulate the thrombin activity. When the concentration of PEP^{WP} was 15 mg mL^{-1}, it could only slow down the increase rate of thrombin activity but could not downregulate the activity of thrombin. In conclusion, PEP^{WP} could inhibit thrombin activity at high concentrations.

3.8 | PEP^{WP} Had Affinity With Thrombin

Surface plasmon resonance (SPR) is a new analytical technique based on optical principles. In this part, SPR was used to further verify the interaction between PEP^{WP} and thrombin.

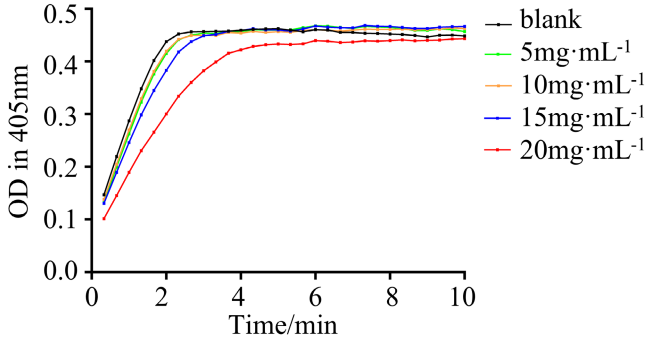


FIGURE 4 | Enzyme kinetics curves of s-2238 chromogenic substrate digested with different concentrations of PEP^{WP} and thrombin.

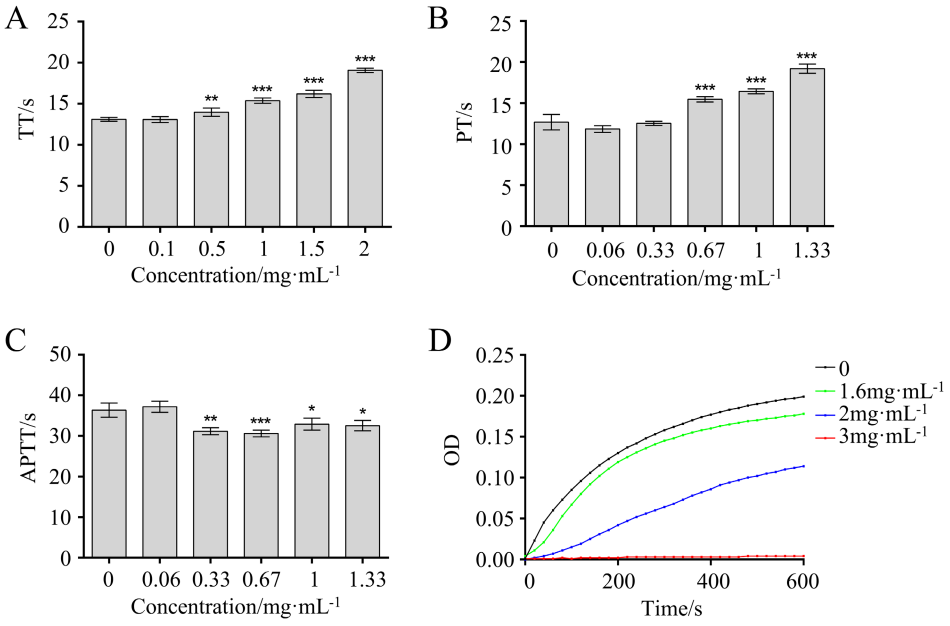


FIGURE 3 | Anticoagulant activity tests of PEP^{WP}. Anticoagulant activity of PEP^{WP} at various of concentrations were expressed with TT (A), PT (B), and APTT (C). The concentrations of PEP^{WP} added were 1, 5, 10 mg, 15, and 20 mg mL^{-1}, and the concentrations converted to the final system are shown. Also, PEP^{WP} could inhibit thrombin-catalyzed coagulation of fibrinogen (D); the concentrations of PEP^{WP} added were 8, 10, and 15 mg mL^{-1}; and the concentrations converted to the final system are shown. *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$, compared to the control group.

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Different concentrations of PEP^{WP} flowed through the surface of the chip with thrombin successively, and the affinity between PEP^{WP} and thrombin was measured. According to the enzyme binding kinetics curve (Figure 5A), the binding curve changed significantly with the increase in the concentration of PEP^{WP}. The affinity parameter (equilibrium dissociation constant [K_d]) of PEP^{WP} and thrombin was 7.242×10^{-6} mol L^{-1} (Figure 5B). For protein-peptide binding, it was generally considered that the affinity parameter of $10^{-7}\sim10^{-4}$ was strong binding, indicating that PEP^{WP} had a strong affinity with thrombin.

3.9 | PEP^{WP} Had Anticoagulant Activity In Vivo

PEP^{WP} was administered by a single tail vein injection. TT, PT, APTT, and fibrinogen time were used to determine the coagulation function of PEP^{WP} (10 mg mL^{-1} administration and 5 mg mL^{-1} administration) in rats under different action times; the results were shown in Table S3 and Figure 6.

In the common pathway, there was a general trend of decreasing clotting time with prolonged action time of PEP^{WP} (Figure 6A,B). The TT of the high-dose group was still different from that of the blank group 60 min after administration, and the TT of the low-dose group was different from that of the blank group 30 min after administration. The change of fibrinogen time was significantly smaller than the change of TT, indicating that the effect of PEP^{WP} on fibrinogen was not significant at the dose concentration set in the experiment (Figure 6G,H). PEP^{WP} did not seem to play a role in the exogenous pathway (Figure 6C,D). PT was not significantly different from those of the blank group at any time after administration regardless of high or low dose. In the endogenous pathway (Figure 6E,F), the APTT of the two groups was similar at all times, indicating that the higher the concentration of PEP^{WP} was not the better when it acted on the endogenous pathway.

Based on the above results, it was concluded that the anticoagulant effect of PEP^{WP} in rats was mainly through the endogenous pathway and the common pathway, and it had an anticoagulant effect when the dose concentration was 5 mg mL^{-1}, and the anticoagulant effect was better when the action time was within 30 min.

3.10 | PEP^{WP} Reduced Blood Viscosity in Acute Blood Stasis Rats

Acute blood stasis in rats was induced by injection of high molecular weight dextran, and different concentrations of PEP^{WP} were injected to evaluate the whole blood viscosity and plasma viscosity of rats in each group. The results are shown in Table S4 and Figure 7. Compared with the blank group, the whole blood viscosity of the model group was about twice that of the blank group, and bivalirudin could also reduce the whole blood viscosity to a certain extent, so it was considered that the modeling was successful. The whole blood viscosity of PEP^{WP} was different from that of the model group only at a low dose (5 mg mL^{-1}), and there was no significant difference at a high dose (10 mg mL^{-1}). There was no significant difference in plasma concentration after any dose, but plasma viscosity tended to decrease after injection of PEP^{WP}. These results indicated that PEP^{WP} had a better effect on reducing blood viscosity when treated at 5 mg mL^{-1} in rats.

3.11 | Mutation Analysis of Thrombin- PEP^{WP} Complex by MD Simulation

The anticoagulant activity of PEP^{WP} was verified by PT, TT, and APTT tests, and PEP^{WP} could inhibit the thrombin-catalyzed coagulation of fibrinogen, suggesting our peptide screening strategy was practicable. The MD structure and 2D interaction map of the Thrombin-PEP^{WP} complex showed that both Arg101 and Arg233 contributed to the binding, and Arg233 was more important because the electrostatic interaction remained through the docking MD simulation. To check the binding mechanism of the complex, we generated the thrombin single mutants Arg101Ala, Arg233Ala, and the double mutant Arg101Ala/Arg233Ala. Specifically, the Arg101 and Arg233 residues of thrombin were substituted with alanine through site-directed mutagenesis.

RMSD of all the complex, including single mutants Arg101Ala, Arg233Ala, and the double mutant Arg101Ala/Arg233Ala, fluctuated dramatically, and the highest RMSD reached 1.5 nm (Figure 8); however, the effect on the RMSD of PEP^{WP} was small (Figure S6). The mutation of Arg101Ala had the fewest large fluctuations, suggesting that the Arg101 residue played a weaker role than the Arg233 residue when PEP^{WP} binding to thrombin. Two-dimensional interaction mapping showed that the mild mutation Arg101Ala did not affect the complex massively because

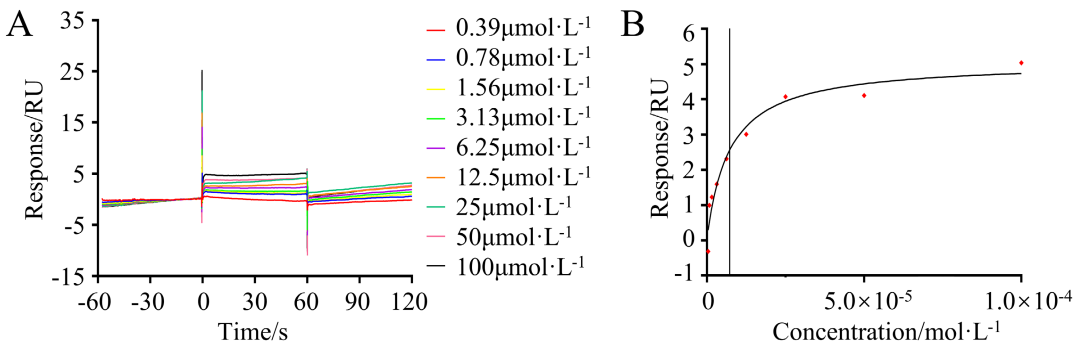


FIGURE 5 | Surface plasmon resonance (SPR) binding curve and steady-state fit plot of PEP^{WP} and thrombin. (A) The binding curve of PEP^{WP} to thrombin. (B) Steady-state fit plot of PEP^{WP} binding to thrombin.

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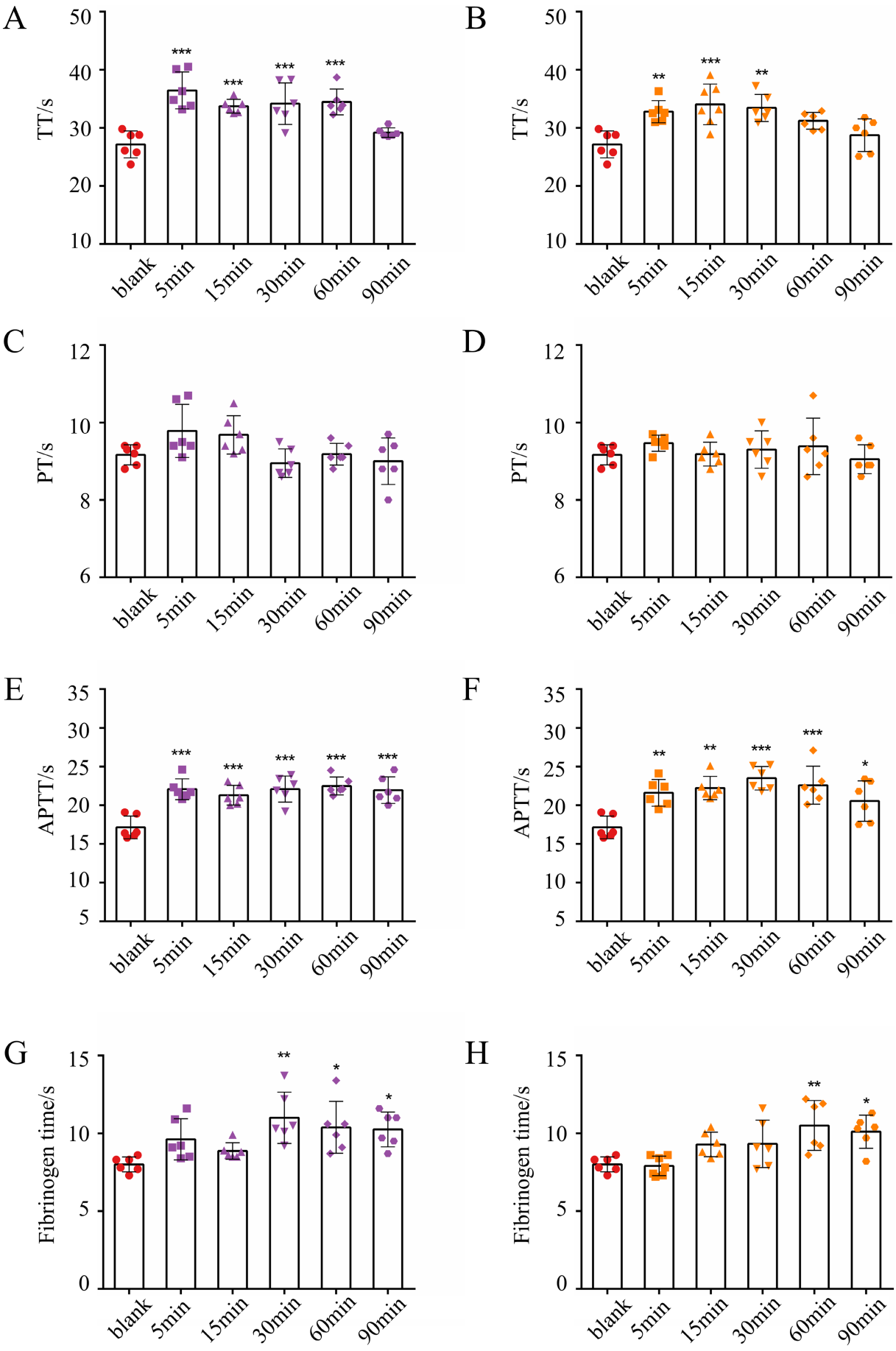


FIGURE 6 | Legend on next page.

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FIGURE 6 | Anticoagulant activity of PEP^{WP} in rats at different time points ($n=6$). Compared with the blank group, $^{*}p<0.05$, $^{**}p<0.01$, and $^{***}p<0.001$. Figures A, C, E and G showed the anticoagulant activity of high-dose PEP^{WP} at different times; Figures B, D, F, and H showed the anticoagulant activity of low-dose PEP^{WP} at different times.

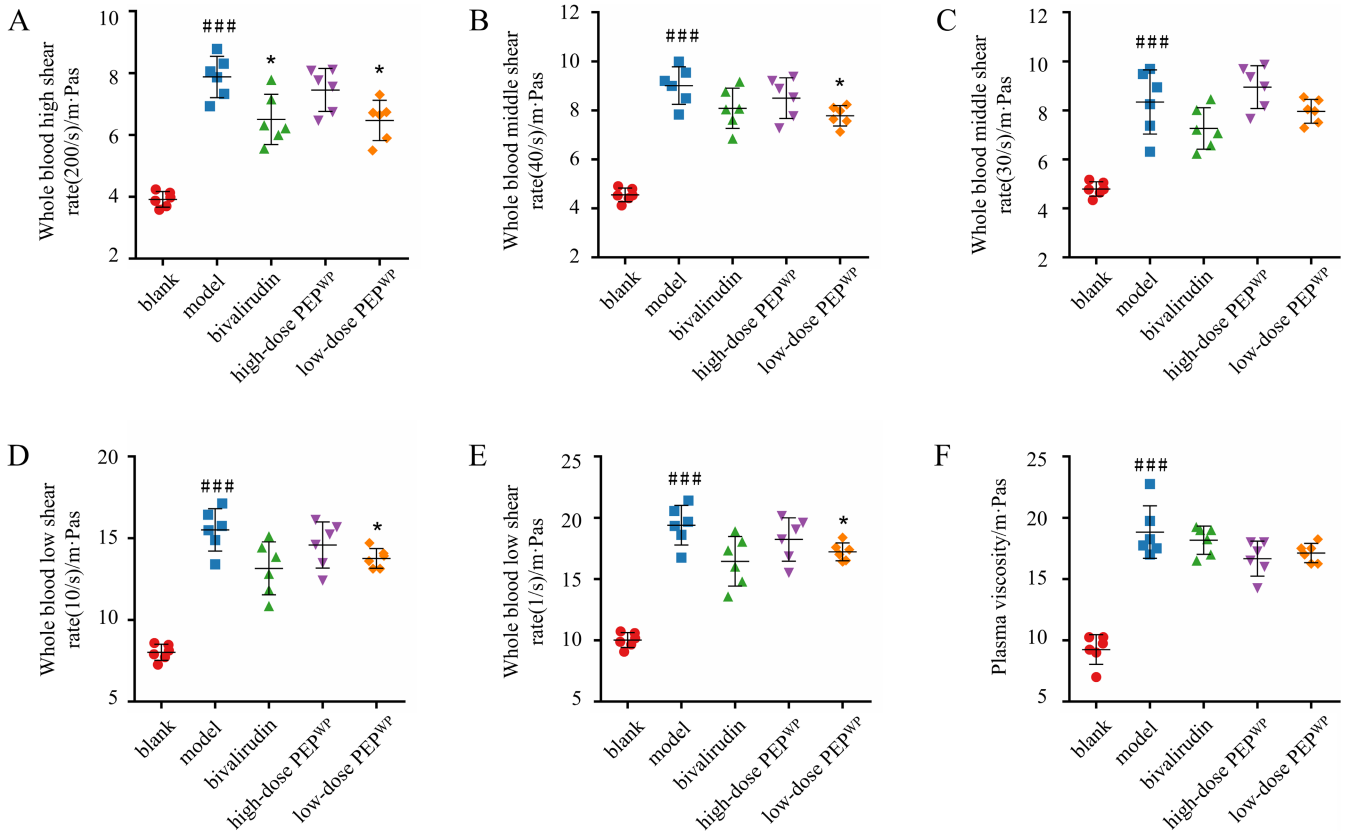


FIGURE 7 | Effect of PEP^{WP} on whole blood and plasma viscosity in rats ($n=6$). Compared with the blank group, $^{###}p<0.001$; compared with the model group, $^{*}p<0.05$.

the binding site was preserved by Arg233 (Figure 9A). However, none of the H-bonds in the wildtype complex were preserved when Arg233 was mutated to alanine, and the electrostatic interaction shifted to Lys236, Lys235, and Arg93 (Figure 9B). The number of H-bonds decreased sequentially in the mutation of Arg101Ala, Arg233Ala, Arg101Ala, and Arg233Ala, and the number was 6, 4, and 2, respectively (Figure 9). The MM/PBSA result of the mutations showed that whether it was a single-site mutation or a double-site mutation, its binding energy was lower than that of the wild type, among which the binding energy of the double-sites mutation was the lowest, only $-64\,kJ\,mol^{-1}$ (Table 3). Taken all these mutation results into consideration, the residue Arg233 was crucial for PEP^{WP} to bind thrombin and activate the anticoagulant function.

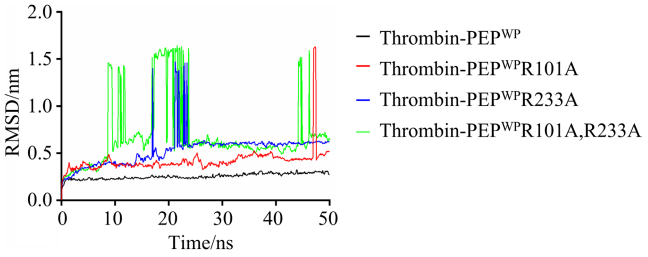


FIGURE 8 | RMSD profiles of the R101A, R233A, R101A, and R233A thrombin-PEP^{WP} complex mutant during 50-ns MD simulation.

4 | Discussion

This paper focused on screening anticoagulant peptides from trypsin hydrolysates of raw and processed WP and investigating the binding interaction between thrombin and the anticoagulant peptide to better understand the anticoagulation mechanism of WP. We identified 160 WP-derived peptides to establish the initial peptide library (Table S1) and then performed molecular docking and MD simulation for two rounds of screening. PEP^{WP} (LRELEDALEQER) was screened out to be the potential anticoagulant peptide. The anticoagulant activity of PEP^{WP} was confirmed by PT and TT tests (Figure 3A,B), indicating that PEP^{WP} can inhibit thrombin in both common and exogenous coagulation pathways. Moreover, APTT test data showed that PEP^{WP} did not inhibit the endogenous coagulation pathway (Figure 3C). Through the chromogenic substrate method and SPR, it was found that PEP^{WP} could bind to thrombin (Figures 4 and 5), and PEP^{WP} could prolong APTT and TT (Figure 6) and reduce blood viscosity in acute blood stasis rats (Figure 7). Additionally,

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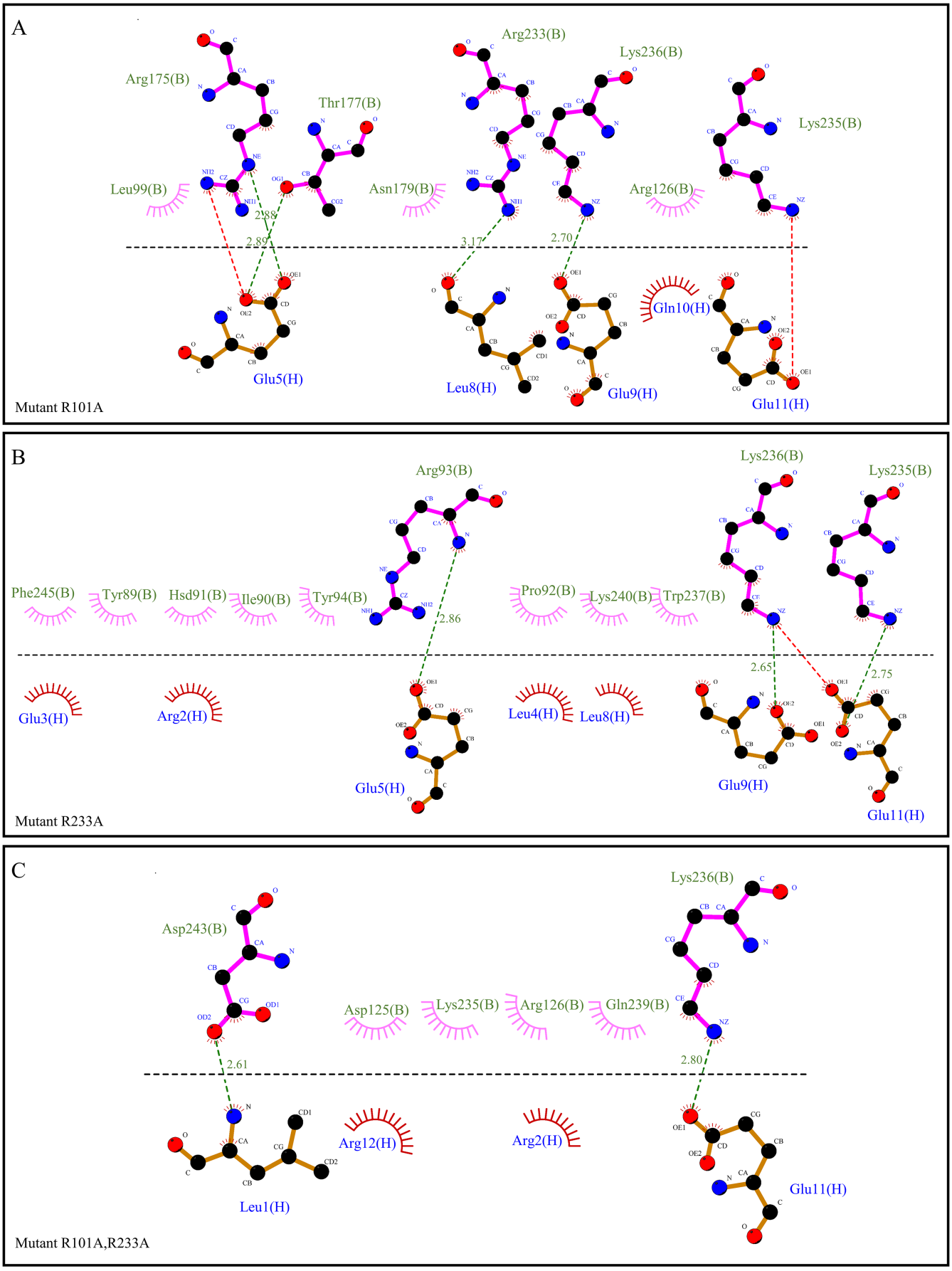


FIGURE 9 | Two-dimensional mapping of the interaction between PEP^{WP} and the R101A thrombin mutant (A), the R233A thrombin mutant (B), and the R101A and R233A thrombin mutant (C) by using Ligplot software.

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TABLE 3 | Results of MM/PBSA of mutation.

Mutation	van der Waals energy (kJ mol^{-1})	Electrostatic energy (kJ mol^{-1})	Polar solvation energy (kJ mol^{-1})	SASA energy (kJ mol^{-1})	Binding energy (kJ mol^{-1})
Wild type (LRELEDALEQER)	−115.103 ± 35.317	−3114.720 ± 494.470	2933.266 ± 441.736	−22.251 ± 4.863	−318.809 ± 120.914
Arg101Ala, Arg233Ala	−23.019 ± 24.670	−1607.979 ± 670.617	1575.117 ± 594.430	−8.270 ± 5.363	−64.151 ± 129.401
Arg101Ala	−74.589 ± 29.509	−2296.228 ± 423.552	2240.255 ± 382.452	−17.340 ± 4.269	−147.902 ± 95.649
Arg233Ala	−105.589 ± 53.575	−2129.455 ± 452.516	2094.320 ± 389.473	−18.430 ± 5.594	−159.154 ± 107.163

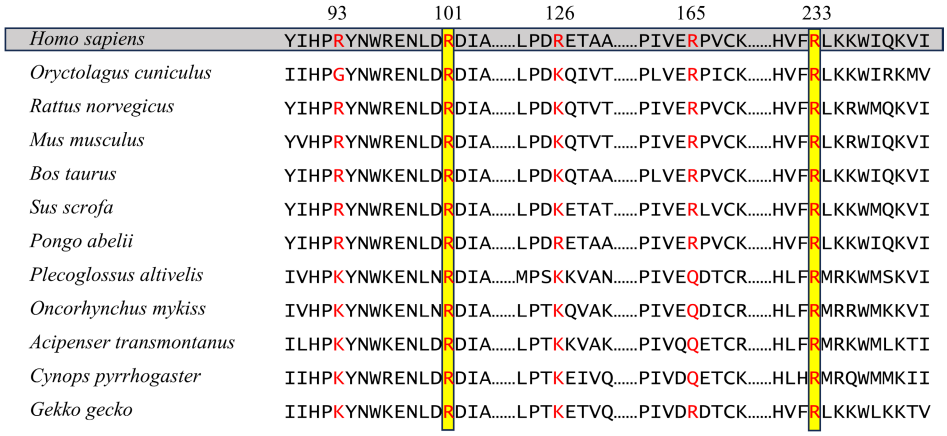


FIGURE 10 | Sequence alignment of the basic residues in Exosite II of thrombin involved of ligand binding (bold in red). Conserved residue alignments were highlighted in yellow.

in contrast to bivalirudin, the dose-dependent inhibition of thrombin activity by PEP^{WP} (Figure 3C) demonstrated its potential for precision dosing adjustments in clinical practice. This pharmacodynamic profile enables individualized dosing regimens based on patient-specific factors (e.g., renal function and thrombotic risk), thereby facilitating optimized anticoagulant management with minimized bleeding risks.

In this paper, more structural details and the interaction between thrombin and PEP^{WP} were found based on the MD simulations. Three-dimensional structure and 2D mapping both showed that Arg101 and Arg233 residues within Exosite II of thrombin serve as primary binding determinants for PEP^{WP} (Figure 2B,C), contrasting with hirudin's Exosite I targeting. Both Exosites I and II exhibit electropositive characteristics, whereas Exosite II prefers to bind ligands with more negative charge. There are five acidic residues in PEP^{WP} (Glu3, Glu5, Asp6, Glu9, and Glu11) that can better match Exosite II. Research proved that allosteric regulation served as the fundamental mechanism for thrombin functional modulation, and the α-helix region of Exosite II acted as a critical allosteric site that mediated cascading amplification effects of thrombin. Our findings demonstrated that PEP^{WP} not only prolonged thrombin time (TT) through binding to thrombin (Figure 3A) but also extended activated partial thromboplastin time (APTT) (Figure 3C) and inhibited fibrinogen polymerization (Figure 3D). These results collectively confirmed that PEP^{WP} binding to Exosite II of thrombin triggered a hierarchical functional suppression cascade in thrombin.

It is well known that hirudin, a potent direct thrombin inhibitor isolated from *Hirudo nipponia* (a blood-sucking leech species), exerts its anticoagulant effect through irreversible binding to Exosite I and the catalytic Ser195 residue of thrombin. However, this strong binding affinity results in significant bleeding liabilities without specific antidotes and limits its clinical applications. In Traditional Chinese Medicine (TCM), WP remains the most clinically utilized leech species. Unlike blood-sucking leeches, WP lacks hirudin and exhibits relatively weak antithrombin activity (Chinese Pharmacopoeia requirement: ≥ 4 U/g vs. ≥ 40 U/g for blood-sucking leeches). Nevertheless, WP has been proved to have significant therapeutic efficacy in anticoagulant and thrombolytic therapies [34, 35]. This unique pharmacological profile confers distinct clinical advantages, particularly manifesting in broader safety margins for therapeutic dosages and enhanced clinical manageability.

In this paper, single-site mutations Arg101Ala and Arg233Ala and mutation both Arg101Ala and Arg233Ala of thrombin-PEP^{WP} complex were performed at last, and the results showed that Arg233 was more crucial for PEP^{WP} to bind thrombin and activate the anticoagulant function. In thrombin-PEP^{WP} complex with the mutation Arg101Ala, the binding interaction of Arg233 was maintained. It is noteworthy that although the binding interface shifted drastically in the Arg233Ala complex, the consequent binding residues on thrombin were changed to Arg93, Lys235, and Lys236 (Figure 9B), which also belong to thrombin Exosite II. An identical set of basic residues Arg93, Arg101, Arg233, Lys235, Lys236, and Lys240 of Exosite II on

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thrombin was essentially used by both unfractionated heparin and GPIbα to bind. Allosteric regulation serves as the fundamental mechanism for thrombin functional modulation. Although nearly all Exosite I-binding ligands rely on specific thrombin residues for interaction, Exosite II binding exhibits no specific residue requirements, which confers broader ligand compatibility and enables a better balance between anticoagulant efficacy and bleeding risk [36]. What is more, we found both Arg101 and Arg233 were conserved across the species (Figure 10). The evolutionary conservation of Arg101 and Arg233 in thrombin not only represents a molecular imprint of thrombin's evolutionary history but also provides a mechanistic rationale for developing Exosite II–targeted anticoagulants. This conservation suggests that inhibitors targeting these residues may achieve broad-spectrum anticoagulation by disrupting conserved functional networks while mitigating preclinical-to-clinical translation risks inherent to species divergence.

5 | Conclusion

In this paper, we constructed a WP-derived peptide library using biomimetic enzymatic hydrolysis, screened out peptide PEP^{WP} (LRELEDALEQER) through molecular simulation, and validated its anticoagulant efficacy through in vitro and in vivo experiments. The inhibitory effect of PEP^{WP} on thrombin activity demonstrated a significant dose-dependent correlation, indicating its potential for precise therapeutic dosing. PEP^{WP} was found to bind to Exosite II of thrombin. The critical residues Arg101 and Arg233 within Exosite II served as the primary binding determinants for PEP^{WP}-thrombin interaction.

Acknowledgments

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Conflicts of Interest

The content of the article has been patented, and the patent application number is 2023118310374. The inventor included Ruijuan Yuan, Peng Wei, Yutong Hua, Ruijuan Dong, Yuting Zhang, Zhaoyuqing Su, and Quancheng Xin.

Data Availability Statement

The datasets used and/or analyzed in the current study are available from the corresponding author on reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Total ion chromatogram (TIC) of Raw WP under condition 1 in positive mode. **Figure S2:** Total ion chromatogram (TIC) of Raw WP under condition 2 in positive mode. **Figure S3:** Total ion chromatogram (TIC) of Processed WP under condition 1 in positive mode. **Figure S4:** Total ion chromatogram (TIC) of Processed WP under condition 2 in positive mode. **Figure S5:** HPLC chromatogram (A) and mass spectrum (B) of PEP^{WP}. **Figure S6:** RMSD profiles of the R101A, R233A, R101A and R233A mutant of PEP^{WP} during 50 ns MD simulation. **Table S1:** Results of thrombin-peptide docking and peptide sources. **Table S2:** Average RMSD of MD simulation. **Table S3:** Anticoagulant activity of PEPWP at different time points in rats ($\overline{x} \pm s$, $n = 6$). **Table S4:** Comparison of whole blood viscosity and plasma viscosity in rats ($\overline{x} \pm s$, $n = 6$).

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